QUALITY INFORMATION DOCUMENT

For SEASTATE significant wave height extreme variability

BLKSEA_OMI_SEASTATE_extreme_var_swh_mean_and_anomaly
IBI_OMI_SEASTATE_extreme_var_swh_mean_and_anomaly
MEDSEA_OMI_SEASTATE_extreme_var_swh_mean_and_anomaly

Issue: 1.2

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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2019/10/17	All	Initial version	Álvaro de Pascual Collar	Marcos García
1.1	2020/04/15		Minor modifications related to the first review of the product	Álvaro de Pascual Collar	
1.2	2025/09/15	All	Update of the upstream products used and the links to documentation. Redrafting of Section 2: Scientific Validation.	Álvaro de Pascual Collar	Copernicus Marine Product Management

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1) INTRODUCTION

a) Product Name

BLKSEA_OMI_SEASTATE_extreme_var_swh_mean_and_anomaly
IBI_OMI_SEASTATE_extreme_var_swh_mean_and_anomaly
MEDSEA_OMI_SEASTATE_extreme_var_swh_mean_and_anomaly

b) Indicator Evaluation Method

The SEASTATE significant wave height extreme variability indicators are based on the computation of the 99th percentile - which represents approximately the highest 1% of significant wave height (SWH) values - derived from model outputs. This approach follows the methodology applied by Woodworth and Blackman (2004) for extreme sea level analysis and is used to detect changes in the frequency distribution of the variable of interest.

All data used for the computation of the extreme variability indicators were obtained from the Copernicus Marine Environment Monitoring Service (CMEMS). The analysis is performed on SWH data for three CMEMS regions: the Black Sea (BLKSEA), Iberia-Biscay-Ireland (IBI), and the Mediterranean Sea (MEDSEA). For each region, the analysis combines information from both Multi-Year Products (MYP) and Near Real-Time (NRT) Analysis Products.

The following table summarizes the CMEMS products used for the analysis for each region.

Black Sea (BLKSEA)	MYP	CMEMS product BLKSEA_MULTIYEAR_WAV_007_006 **PUM:** https://documentation.marine.copernicus.eu/PUM/CMEMS-BLK-PUM-007-006.pdf **QUID:** https://documentation.marine.copernicus.eu/QUID/CMEMS-BLK-QUID-007-006.pdf
	NRT	CMEMS product BLKSEA_ANALYSISFORECAST_WAV_007_003 **PUM:** https://documentation.marine.copernicus.eu/PUM/CMEMS-BLK-PUM-007-003.pdf **QUID:** https://documentation.marine.copernicus.eu/QUID/CMEMS-BLK-QUID-007-006.pdf
Iberia-Biscay-Ireland	MYP	CMEMS product IBI_MULTIYEAR_WAV_005_006 **PUM:** https://documentation.marine.copernicus.eu/PUM/CMEMS-

(IBI)		IBI-PUM-005-006.pdf **QUID:** https://documentation.marine.copernicus.eu/QUID/CMEMS-IBI-QUID-005-006.pdf
	NRT	CMEMS product IBI_ANALYSISFORECAST_WAV_005_005 **PUM:** https://documentation.marine.copernicus.eu/PUM/CMEMS-IBI-PUM-005-005.pdf **QUID:** https://documentation.marine.copernicus.eu/QUID/CMEMS-IBI-QUID-005-005.pdf
Mediterranean Sea (MEDSEA)	MYP	CMEMS product MEDSEA_MULTIYEAR_WAV_006_012 **PUM:** https://documentation.marine.copernicus.eu/PUM/CMEMS-MED-PUM-006-012.pdf **QUID:** https://documentation.marine.copernicus.eu/QUID/CMEMS-MED-QUID-006-012.pdf
	NRT	CMEMS product MEDSEA_ANALYSISFORECAST_WAV_006_017 **PUM:** https://documentation.marine.copernicus.eu/PUM/CMEMS-MED-PUM-006-017.pdf **QUID:** https://documentation.marine.copernicus.eu/QUID/CMEMS-MED-QUID-006-017.pdf

The first step of the methodology consists of computing the mean annual 99th percentile for each MYP product. This is calculated as the average of the yearly 99th percentiles over the entire time series. In the second step, the same calculation is performed for the most recent year available in the NRT product. The resulting value is then compared with the MYP mean by subtraction, yielding the anomaly map of the 99th percentile for the latest year (**Figure 1**).

Iberia-Biscay-Ireland wave height extreme variability mean and anomaly.

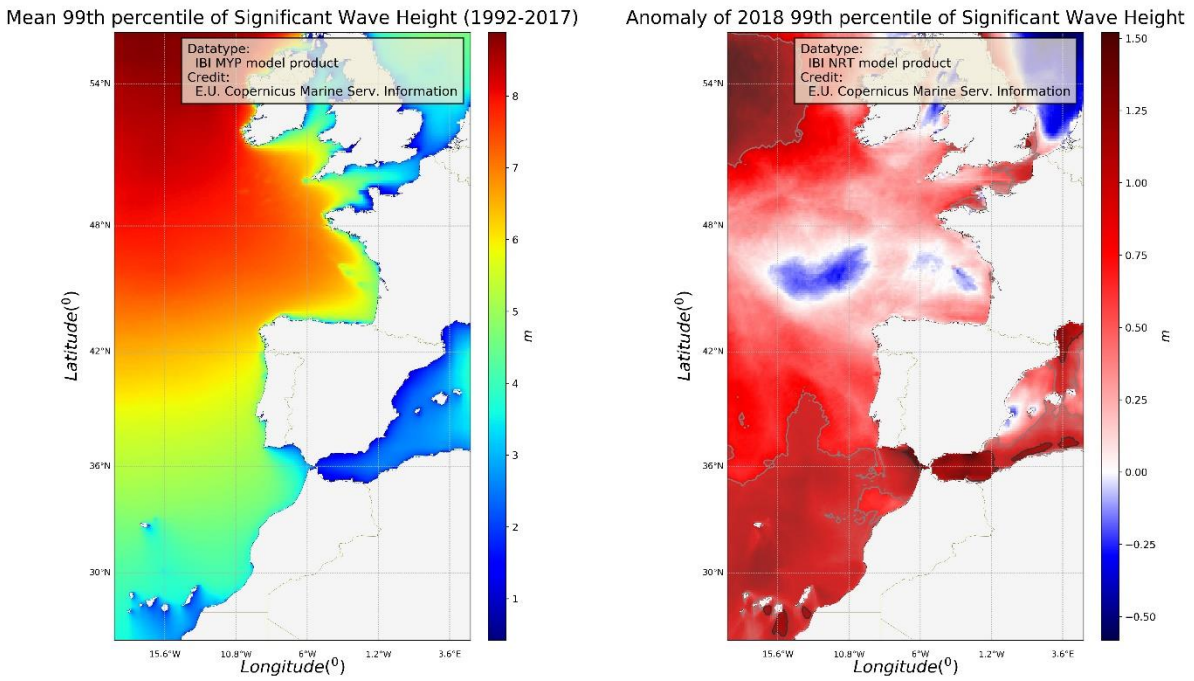

Figure 1. Map of the 99th mean percentile computed from the Multi Year Product (left panel) and anomaly of the 99th percentile in 2018 computed from the Analysis product (right panel). Transparent grey areas (if any) represent regions where anomaly exceeds the climatic standard deviation (light grey) and twice the climatic standard deviation (dark grey).

More details, including the consistency with the same OMI but computed from in situ observations (moored buoys), can be found in the CMEMS Ocean State Report (Pérez Gómez et al., 2016; Pérez Gómez et al., 2018; Alvarez Fanjul et al., 2019).

2) SCIENTIFIC VALIDATION

The validation of this product has been carried out by comparing the results obtained from modelling products with in situ observations (Álvarez Fanjul et al., 2019). This comparison shows a variable degree of consistency across regions, with some discrepancies attributable to local limitations of the models or of the observational time series.

Figure 2 provides an integrated view of extreme wave conditions, combining model-based estimates with buoy records for the reference period 2000–2016 and for the year 2017. It serves as the main framework for evaluating the agreement between simulations and observations, and for analysing the regional patterns and anomalies of significant wave height extremes.

In the IBI region, the model results capture well the latitudinal pattern of extreme wave heights, with 99th percentiles exceeding 8 m in the northern Atlantic (**Figure 2a**). The buoys show good consistency with the models, except at shallow coastal stations such as the Gijón buoy, where local effects are not properly resolved by the modelling system. For 2017, both models and observations agreed in showing positive anomalies across most of the domain and negative values west of Ireland (**Figure 2b**). However, along the Cantabrian coast some buoys registered opposite anomalies, most likely due to data gaps during severe storm events.

In the Mediterranean, the models reproduce the location of the largest extreme values in the Gulf of Lion, the Balearic Sea and southwest Sardinia, associated with Mistral and Tramontana winds, as well as in the Aegean and Levantine Seas, where Bora and Etesian winds prevail (**Figure 2c**). Comparison with in situ buoy data confirms this distribution. In 2017, both model simulations and buoy observations indicated positive anomalies in the western basin and negative anomalies in the eastern basin (**Figure 2d**), consistent with stronger-than-usual winds in the west and weaker conditions in the east.

In the Black Sea, the analysis relied exclusively on model results due to the lack of adequate observational records. Nevertheless, the 2017 simulations showed good agreement with satellite altimeter data (Jason-3 and Sentinel-3a) and with the Varna buoy, reproducing an unusual pattern with maxima shifted towards the central and eastern parts of the basin (**Figure 2e-f**). Anomalies reached values of up to +0.7 m in these areas, in contrast with the negative values observed in the western part of the basin.

More details and a full scientific evaluation can be found in the CMEMS Ocean State Report (Álvarez Fanjul et al., 2019; Pérez Gómez et al., 2018).

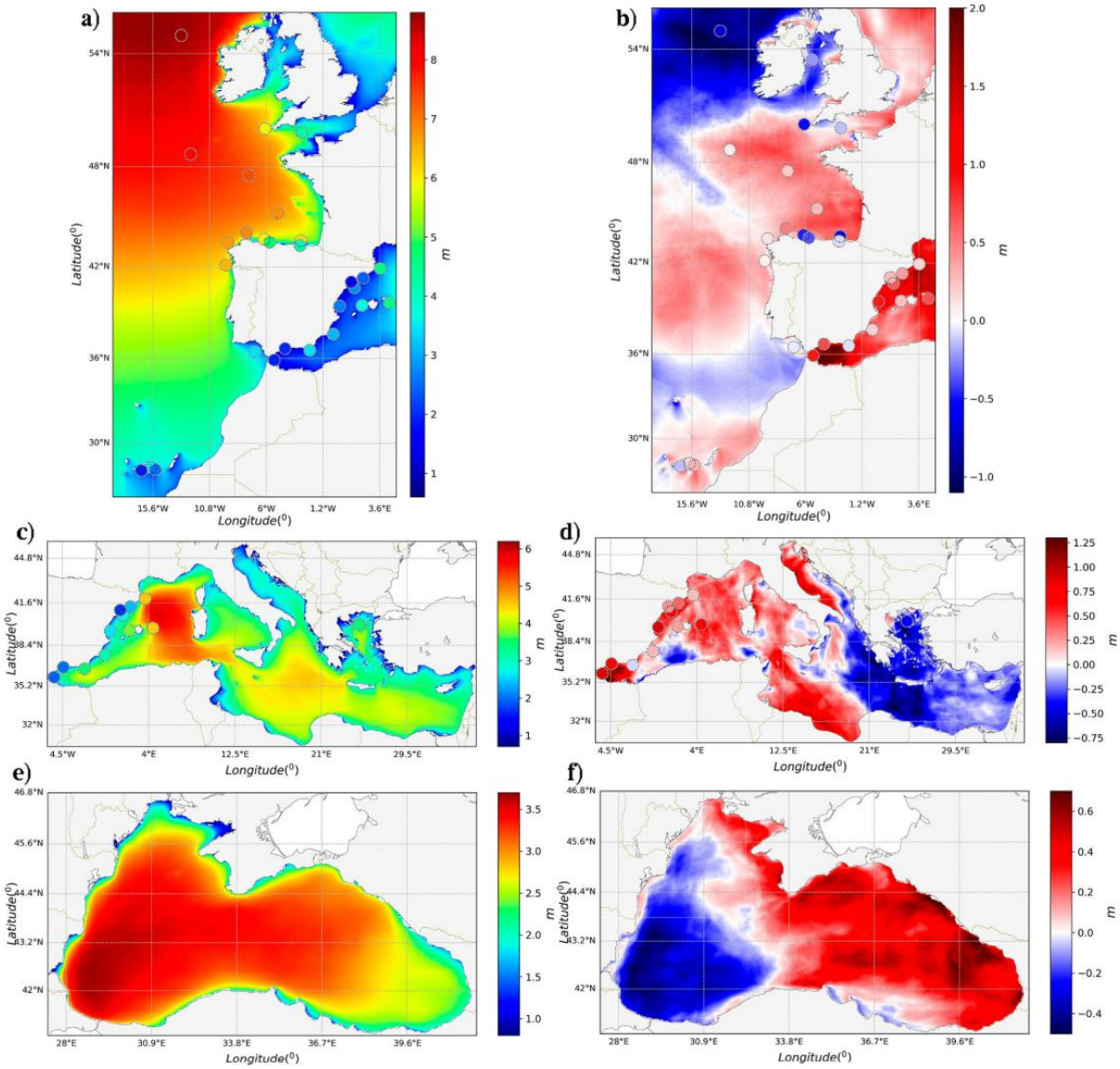

Figure 2. *Spatial distribution of the mean 99th percentile of significant wave height (SWH) computed from CMEMS reanalysis and regional model products together with in situ buoy data (left panels), and the corresponding anomalies for 2017 relative to the 2000–2016 reference climatology (right panels). The figure illustrates the Iberia–Biscay–Ireland (a–b), Mediterranean (c–d), and Black Sea (e–f) regions. Adapted from Álvarez Fanjul et al. (2019).*

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